

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 20

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

3/8/2026

CSA1507 - CLOUD COMPUTING

K. Omkar Grewar

192311431

CO2 - AT2

1) Introduction :- Software Defined Storage (SDS) is a storage Architecture that separates storage management software from the Physical storage hardware, It allows organisation to manage storage Resources through software instead of relying on dedicated storage devices.

→ Easy Expansion :- New storage devices can be added without replacing the Existing Infrastructure.

→ Horizontal Scaling :- Multiple storage servers can be combined into a single storage Pool.

→ centralized Management :- All stored Resources are managed from a single software interface.

→ cloud Integration :- SDS supports Private, Public and Also Hybrid cloud Environments.

Advantages :-

→ High scalability

→ Lower Operational cost

→ Better Storage utilization

→ Easy management.

Conclusion :- Software Defined storage significantly improves scalability by allowing Easy Expansion of storage Resources and Enhances Resource utilization through storage Pooling;

Automation and dynamic Allocation.

2) Introduction :-

Software Defined Networking (SDN) is a networking approach that separates the network control plane from the data forwarding plane. The network is managed centrally using software controllers.

→ centralized Network Management :- All network devices are controlled from a central SDN controller.

→ Automation :- Network configuration and management tasks are automated.

→ simplified Network configuration :- changes can be made through software instead of configuring each switch (or) router individually.

→ Dynamic Traffic Management :- SDN automatically adjusts traffic flow based on network conditions.

→ Easy Troubleshooting :- centralized monitoring helps quickly detect and resolve network issues.

→ Vendor Independence :- SDN supports devices from the multiple vendors.

Conclusion :- Software - Defined Networking simplifies network operations through centralized management, automation and programmable control.